

E-Commerce Intelligence: From Data to Insight

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ABOUT THE DATA

E-commerce dataset containing records about user behavior, item relevance, shopping patterns, etc.

12,000

rows - one per user interaction

47 → 30

raw to cleaned and columns

0

missing value and duplicates found

Source: Kaggle - "E-Commerce Intelligence Data" (*likely synthetic)

17 recommender-algorithm output columns (e.g. embedding similarity, Q-value scores) were dropped as out of scope. Because the dataset is synthetic, this analysis is best read as a demonstration of method, not a real-world market finding.

APPROACH

From raw file to insight: the pipeline

1. Extract

load the raw
[12,000 ,47] .csv

2. Clean

Drop out of scope
columns, check
nulls/duplicates

3. Design & Load

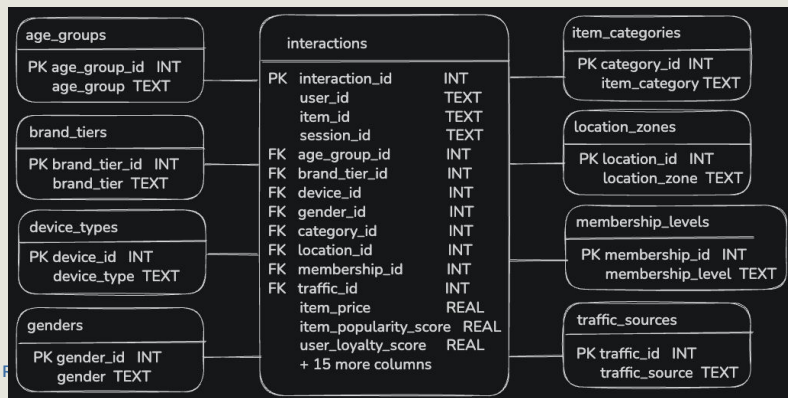
1 fact + 8 look up
tables

4. Query

8 SQL queries, to
answer the RQ

5. Report

Charts & Findings
per RQ



Schema Design:

One interactions fact table, with 8 small lookup tables for repeated categorical variables (eg: age group, category, membership level, etc.)

Who the customer is and where or how they shop does not predict purchase.

RQ1: Does who the customer is predict purchase?

Variables: Age group, Gender, Membership Level, Location Zone

No

All four variables showed very weak association
Cramer's $V \leq 0.025$ with purchase

RQ2: Does where and how they shop predict purchase?

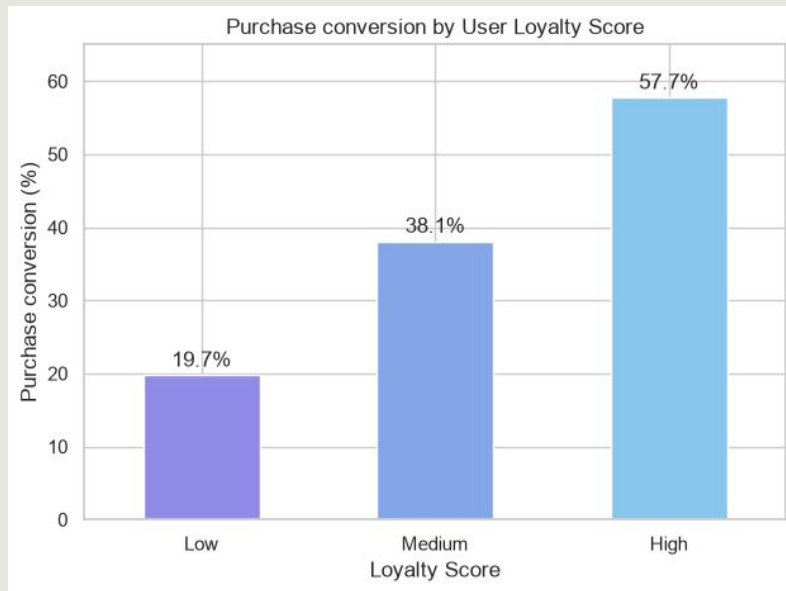
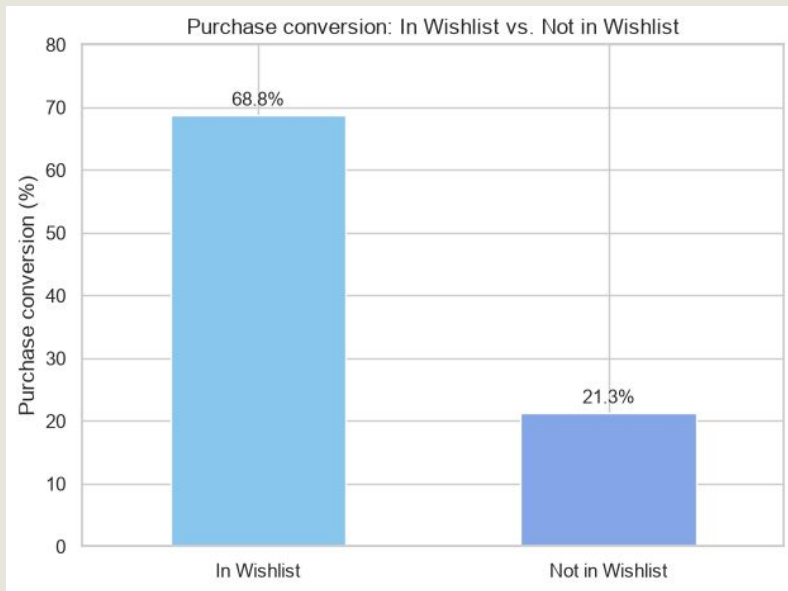
Variables: Item Category, Brand Tier, Device Type, Traffic Source

No

All four variables showed very weak association
Cramer's $V \leq 0.025$ with purchase

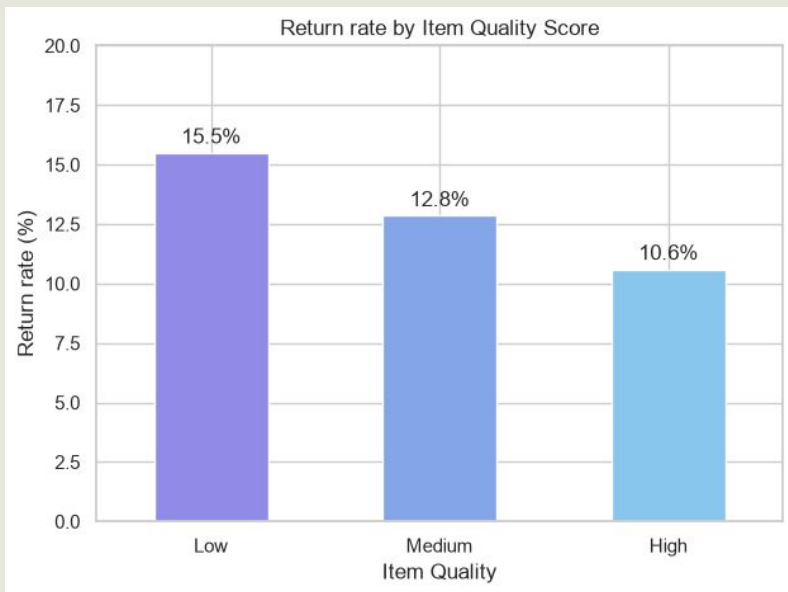
What actually predicts a purchase?

Behavioral intent signals predict purchases. Not who the customer is, not where they shop.



Wishlist status is the strongest signal found anywhere in the analysis (3.2x conversion rate difference). Loyalty score follows a clean, monotonic pattern. Cart-add count, rating, and item quality/popularity show smaller, consistent effects in the same direction.

Among purchases, what predicts a return?



Variables: Item Quality, Customer Rating, Discount Percentage, Item Price

Item quality shows a modest and consistent association with returns however, it's not a very strong one.

Customer rating showed **slight patterns associated** with returns, although not a very strong one.

The rest of the variables showed **no meaningful** relationship with returns.

Limitations and next steps

Limitations:

- The dataset is synthetic, so absolute numbers (37% baseline conversion) are not real-world benchmarks
- The findings demonstrate the analysis method only, not a real shopper behavior

Next Steps:

- Explore stock availability vs. purchase - some categories may be out of stock more often, leading to lower conversion
- Combine click count, engagement score, and loyalty score for a fuller view of shopping behavior

Thank you!